

Evaluating Spectral Interpretability in Brazilian Portuguese Audio Deepfakes via Causal Occlusion

Como as pistas acústicas de um detector XLS-R mudam entre zero-shot e adaptação leve — TODO:
confirmar/traduzir

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ABSTRACT

TODO: abstract.

KEYWORDS

audio deepfake detection, anti-spoofing, explainability, Spearman correlation, spectral occlusion, MFCC, cross-lingual, wav2vec 2.0, XLS-R, Brazilian Portuguese

1 INTRODUCTION

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We frame the study around one research question: *which acoustic cues does the detector rely on, and how do these cues shift after a light adaptation to Brazilian Portuguese?* We evaluate two detectors sharing a frozen XLS-R front-end: a zero-shot detector (D_{zs}) and a lightly adapted one (D_{ad}).

One condition must hold for the comparison to be meaningful: adaptation must improve detection. We require $EER(D_{ad}) < EER(D_{zs})$ before any cue analysis; otherwise the “before/after” contrast is reported as *insufficient adaptation*. Because our associative analysis is a direct, model-agnostic rank correlation (rather than attributions from a fitted surrogate), it carries no separate surrogate-fidelity precondition.

We investigate three hypotheses:

- **H1 (associative cue shift).** The acoustic-association profile over MFCC features differs between D_{zs} and D_{ad} . We measure association as the *intra-class Spearman rank correlation* between each MFCC and $P(\text{spoof})$ (FDR-controlled per detector), and compare the profiles *descriptively* across detectors; we make no significance claim on the between-detector contrast.
- **H2 (causal spectral shift).** The frequency bands whose occlusion reduces $P(\text{spoof})$ differ between D_{zs} and D_{ad} . Because occlusion is an intervention on the input, we report the resulting causal profiles *with quantified uncertainty* (95% bootstrap confidence intervals on the per-band mean drop): a band whose interval excludes zero exerts a reliable causal effect on the score. We stop short of a formal significance test across detectors.
- **H3 (error-linked acoustic differences).** For the MFCC features most salient by intra-class Spearman $|\rho|$, distributions differ across decision quadrants: in mean between true-negative and false-positive clips (Welch’s t -test) and in variance between true-positive and false-negative clips (Levene’s

test). H3 is tested *inferentially*, controlling the false discovery rate (Benjamini–Hochberg, $q < 0.05$).

2 RELATED WORK

2.1 Self-Supervised Front-Ends for Anti-Spoofing

Tak et al. [4] TODO.

2.2 Explainability in Audio Deepfake Detection

Grinberg et al. [3] and Fullwood and Monroe [2] TODO.

2.3 Cross-Lingual Generalization and Real-World Deployment

Channing et al. [1] TODO.

3 METHODOLOGY

3.1 Detectors Under Study

TODO.

3.2 Pre-processing

TODO.

3.3 Acoustic Features (MFCC)

TODO.

3.4 Spine 1 — Acoustic Association (Spearman)

TODO.

3.5 Spine 2 — Spectral Causal Occlusion

TODO.

3.6 Confirmatory Statistics

TODO.

4 EXPERIMENTAL SETUP

Dataset and sampling rationale. BRSpeech-DF is a large-scale Brazilian-Portuguese anti-spoofing corpus (459,137 clips: 76,644 bona fide and 382,493 spoofed, the latter carrying a generator-method attribute), split into train/validation/test. Its test partition alone holds 1,320 bona fide and 6,587 spoofed clips. Because our objective is population-level acoustic *explainability* rather than

Table 1: Experimental configuration.

Parameter	Value
Sample rate	16,000 Hz
Fixed window	64,600 samples (≈ 4.04 s)
Front-end	wav2vec 2.0 / XLS-R (frozen)
Checkpoint	nii-yamagishilab/mms-300m-anti-deepfake
Adaptation head (D_{ad})	logistic regression on 1024-d embeddings
Association measure	intra-class Spearman ρ (MFCC vs. $P(\text{spoo})$
MFCC features	13 coeff. $\times$ (mean, std) = 26
Dataset	AKCIT-Deepfake/BRSpeech-DF
Corpus scale	459,137 clips (76,644 bona fide / 382,493 sp)
Train per class (D_{ad} head)	1,500 (balanced)
Analysis set (test)	1,320 bona fide + 1,500 spoof
Positive class	spoof
Seed	42

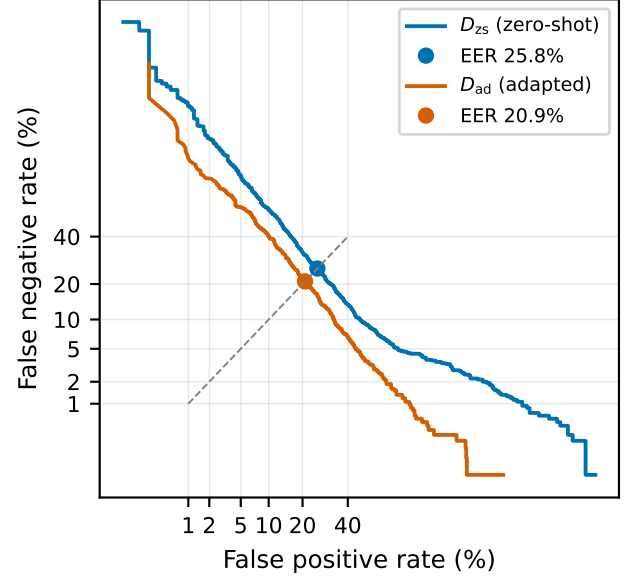
Table 2: Master table schema (one row per audio clip; $_{zs}$ = zero-shot, $_{ad}$ = adapted).

Column	Description
sample_id	unique clip identifier
ground_truth	0 = bonafide, 1 = spoof
spoof_algorithm	generator/attack (null if absent)
p_spoof_{zs,ad}	continuous $P(\text{spoof})$ per detector
pred_{zs,ad}	decision at EER threshold
quadrant_{zs,ad}	TP / TN / FP / FN
mfcc{i}_mean	mean of i -th MFCC ($i = 1..13$)
mfcc{i}_std	std of i -th MFCC ($i = 1..13$)

leaderboard-style benchmarking, we deliberately analyse a compact, class-controlled subset instead of the full corpus. We take the *entire* bona fide test split (1,320 clips — the scarce class) matched with 1,500 spoofed test clips, and fit the D_{ad} logistic head on an independent, class-balanced training subset (1,500 per class); test clips are never seen during adaptation, precluding leakage. This scale is a design choice, not a data limitation. First, every inferential claim (H3: Welch and Levene under Benjamini-Hochberg control) and every interval estimate (H2: per-band bootstrap CIs) is already well powered at this N : at the EER operating point the rare error quadrants (FP, FN) still retain a few hundred clips each, enough to resolve small-to-moderate effects ($d \approx 0.25\text{--}0.3$ at 80% power), so additional clips would add negligible statistical resolution. Second, exhausting the scarce bona fide class while keeping a comparable spoof sample keeps the four decision quadrants (TP/TN/FP/FN) and the EER threshold meaningful, which the corpus's native $\approx 1:5$ bona fide/spoof imbalance would otherwise distort. Third, the per-clip analysis (band-stop spectral occlusion with a per-band bootstrap) scales with N , so a focused subset preserves full reproducibility at tractable cost. Scaling the analysis to the full corpus and stratifying it by spoof generator is left to future work.

Table 3: Detection performance on the analysis set (performance_table).

Detector	EER	MCC	Accuracy
D_{zs} (zero-shot)	25.8%	0.483	0.742
D_{ad} (adapted)	20.9%	0.581	0.791

**Figure 1: Detection error trade-off (DET) curves for D_{zs} and D_{ad} on the analysis set; the marker on each curve is the operating point at the Equal Error Rate. A curve closer to the origin indicates better detection.**

5 RESULTS

5.1 Zero-Shot vs. Adapted Performance

Figure 1 contrasts the two detectors; the EER pre-condition holds when the D_{ad} curve lies below D_{zs} . TODO.

6 EXPLAINABILITY ANALYSIS

6.1 Acoustic Association via Spearman

Figure 2 compares the per-feature association profiles of D_{zs} and D_{ad} . TODO.

6.2 Spectral Occlusion Profiles

Figures 3 and 4 report the causal spectral profiles. TODO.

6.3 Confirmatory Tests

Figure 5 shows the quadrant-wise distributions with FDR-corrected significance. TODO.

As a complementary associative view, Figure 6 relates the most correlated MFCC to the detector score within each class (spearman_table). TODO.

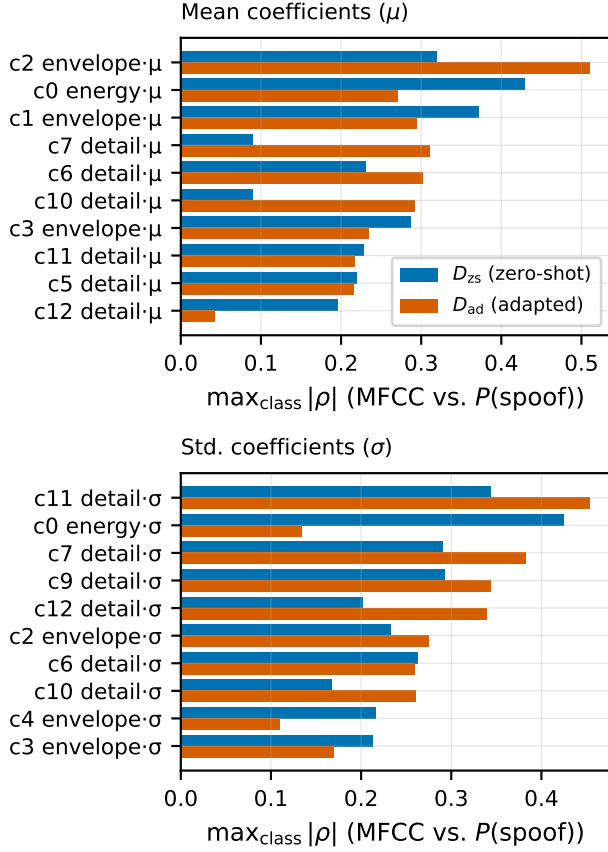


Figure 2: Acoustic-association profile of the MFCC features, measured as the intra-class Spearman $|\rho|$ between each coefficient and $P(\text{spoof})$ (maximum over the bona fide/spoof classes), split into mean (μ) and standard-deviation (σ) coefficients. Labels follow cepstral order: c_0 (energy), c_1 – c_4 (envelope/tilt), c_5 – c_{12} (fine detail). We read the between-detector shift *descriptively* (H1).

7 DISCUSSION AND LIMITATIONS

TODO.

8 CONCLUSION AND FUTURE WORK

TODO.

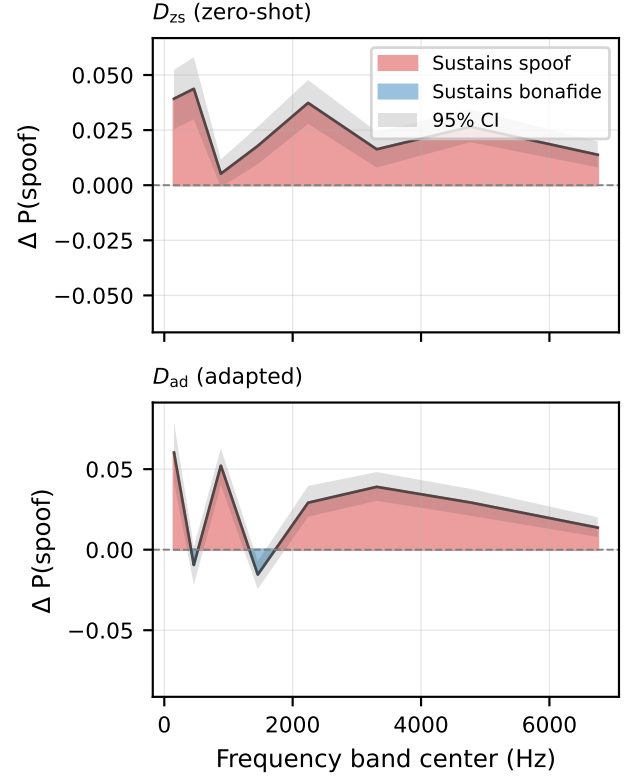


Figure 3: Divergent spectral occlusion profiles (D_{zs} top, D_{ad} bottom). Red area: occluding the band lowers $P(\text{spoof})$ (the band *sustains spoof*); blue: occlusion raises it (sustains *bonafide*); grey ribbon: 95% bootstrap CI of the per-band mean drop. Bands whose CI excludes zero exert a reliable causal effect (H2).

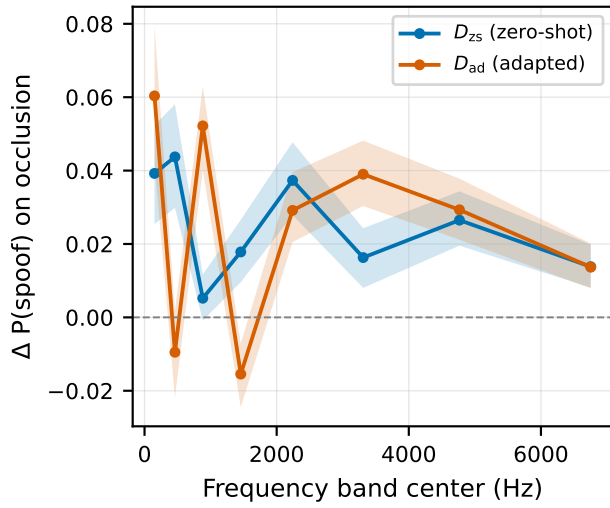


Figure 4: Direct overlay of the two mean occlusion profiles with 95% bootstrap CI ribbons, easing the band-by-band D_{ZS} vs. D_{ad} comparison.

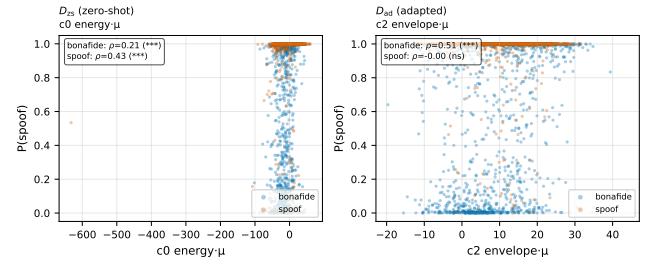


Figure 6: Intra-class association between the most correlated MFCC and $P(\text{spoof})$ for each detector. Points are coloured by ground-truth class; Spearman's ρ (with post-FDR significance) is reported per class. Computing ρ within each class avoids the spurious correlation induced by class separation.

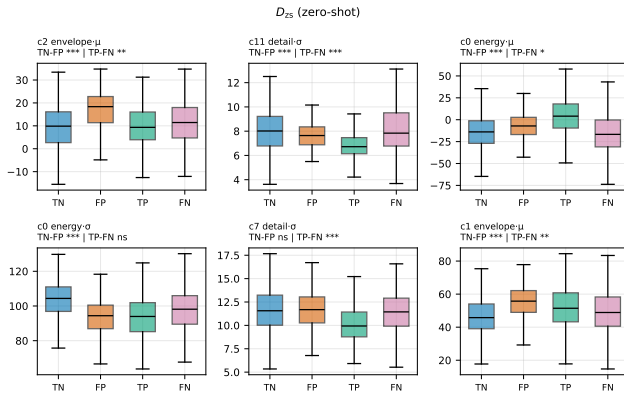


Figure 5: Distribution of the most association-salient (Spearman $|\rho|$) MFCC features across decision quadrants for D_{ZS} . Per-panel annotations give the post-FDR significance of the mean contrast (TN vs. FP, Welch) and the variance contrast (TP vs. FN, Levene): *** $q < 0.001$, ** $q < 0.01$, * $q < 0.05$, ns otherwise (H3, inferential). The analogous figure for D_{ad} is `confirmatory_box_ad`.

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